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PHILADELPHIA, JULY, 1890.

NO. 7

THOS. H. McCOLLIN, Managing Editor.

J. F. SACHSE, Associate Editor.

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AMERICAN JOURNAL OF PHOTOGRAPHY, VOL. XI, NO. 7, JULY
1890 ***

THE AMERICAN JOURNAL OF PHOTOGRAPHY.



"FOLWELL'S WASHINGTON."

NEGATIVE FROM ORIGINAL. BY JULIUS F.
SACHSE.

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THE SILHOUETTE.

Our illustration, "Folwell's Washington," is a profile of the one person characterized in our nation's history as the "First in war, the first in peace, and the first in the hearts of his countrymen." Our object in presenting this frontispiece to our readers for the current month is a two-fold one;--first, in view of the Eleventh Annual Convention of the Photographic Association of America, which is to be held in Washington, August 12--15. The subject is a particularly appropriate one.

The original portrait was painted by Folwell, in 1795, while General Washington was in the presidential chair, for Col. William Washington, a kinsman of the General, and who in the year 1800, but a short time after the General's death, presented the portrait to James Henry Stevens.

The following endorsement is written on the back of the picture: "Done 1795,--Presented to--James Henry Stevens, Esqr.,--by his friend Col.--Wm. Washington, Sept.--9th, 1800--Said to be a--Correct likeness from life of--His Excellency Gen'l--George Washington--1st President of--the United--States of America."

The original is now in the possession of the Historical Society of Pennsylvania, and is classed among what are known as "rare Washington portraits." In Mr. Wm. S. Baker's list we find on page 109 the following notice regarding the portrait and the painter. "Samuel Folwell, 1795, miniature painter, of whom little is known, was practicing his art in Philadelphia, the latter part of last, and the beginning of the present century. The profile of Washington in possession of the Historical Society of Pennsylvania, inscribed 'S. Folwell, Pinxt, 1795,' is said to have been taken upon a public occasion, the President being unaware of the fact. It is drawn on paper and solidly painted in India ink, with certain lights touched in, and

as declared at the time is ‘certainly a most spirited and correct likeness.’ There is no engraving of this profile.”

In addition to this portrait by Folwell, there are in existence two regular silhouettes of Washington. One was taken by Samuel Powell, an ex-mayor of Philadelphia, by tracing on the wall a shadow thrown by an Argand lamp, which had just then been invented. This picture is now in the possession of the Massachusetts Historical Society. The other, a psaligraph, or silhouette cut with the scissors, by a Miss De Hart, of Elizabethtown, N. J., in 1783.

It will be noticed that the portrait belongs to the more advanced type of its class; viz.: where an attempt is made to introduce detail by the use of the brush.

Our other reason for reproducing this portrait is owing to the fact that the solid profile has of late again attracted considerable attention in photographic circles of Europe, and strange to say much difficulty seems to have been experienced in obtaining desirable results. As simple as the production of a solid profile seems to be (and is to us in America)--if we may credit our European exchanges--just so difficult has the process proved to some of the camera artists beyond the sea, as will be seen from the accounts here given.

On account of the extreme plainness and simplicity of the old silhouettes, we are too apt to pass them by as productions of but little or no artistic merit or value, overlooking the fact that the plain outline contour, in centuries past, was the germ from which emanated the arts of bas-relief and portraiture, the latter passing through all the various stages from the plain outline of the finished miniature, and these in their turn to be supplanted by photography of the present day.

There are two traditions which have come down to us from the dim ages past relating to this subject. One informs us that it occurred to an old Greek to follow the contour of a friend’s shadow with a piece of coal as it fell upon the white marble wall of the temple, thus permanently securing the outline of his friend’s features; from this incident is said to date the Greek School of Painting. Arkides, of Corinth, and Telephanes, of Sikyon, improved the process by filling out the space between the lines with a piece of coal or ruddle (an argillaceous iron ore), from thence the transition to pigments was an easy and natural one.

The other tradition, a still more pleasant one, would have us believe that about twenty-five hundred years ago there lived in same city of Sikyon, in Greece, a modeller in clay by the name of Dibutades. He had a daughter Kora; he also had a young apprentice. As usual in these old tales, both were young and fair, and in the course of events vows of betrothal were exchanged. Shortly afterwards, as the pair sat together, Kora suddenly seized a piece of coal from the brazier, and asking her betrothed to remain still, she traced upon the wall the outlines of the face which was so dear to her. It was an inspiration on the part of the Greek maiden, and so correct was the likeness that when old Dibutades saw it he recognized it at once, and thinking to please his daughter, he filled in the portrait with clay; the result was a bas-relief, the first that was ever made.

As crude as the silhouette appears, it certainly was the best process, prior to the advent of photography, to reproduce the features of persons; this applies, however, more to such as had a marked or prominent profile, the result almost always being a recognizable portrait, while in subjects with soft harmonic lines, especially female faces, the identification of the original by means of the silhouette, or profile, was more or less difficult. Yet there was a time, we will say in the century preceding the perfection of Daguerre's invention, when the silhouette was popular and common as the carte or cabinet photo of the present day.

It was about the middle of last century when the rage for profiles broke out in France. It is said the style was introduced by the Pompadour, then at the zenith of her power. Be that as it may, it struck the popular fancy, as it was a branch of portraiture which came within the reach of all classes. Paris was soon flooded with profile artists, and the black profiles became known as "à la Pompadour." With the decline of her power, and the appointment of one Etienne de Silhouette as Minister of Finance, who on account of a system of retrenchment inaugurated by him had become an object of derision with the court favorites and the populace, a reaction set in; and so great a butt had the Minister become with the populace, that everything that savored of retrenchment, or was cheap, poor, or shopworn, was "à la Silhouette." Our profiles, on account of their inexpensiveness, came under the same category, and strange to say thus immortalized, for all time to come, the name of the honest Minister of Finance.

Towards the end of the last century the art of profile painting reached its highest development, of which our example is a good specimen. In our own country, the demand for these pictures was so great that a special machine was invented for the purpose of producing correct outlines in miniature. This apparatus was one of the features of Peale's Museum, then in Independence Hall. The process was first to outline the shadow, then the machine was brought into play; this consisted of a tracer, which moved on a universal joint on the standard, the respective ends being adjustable as to length, so as to suit the required relative proportion between the shadow and the miniature copy. In using, one end of the tracer was caused to follow the line of the profile, while the other marked upon the paper which was presented in a frame. The paper was then removed, and the portrait cut out by the scissors. The silhouette portrait also came into vogue for book illustrations, and specimens from copperplates are frequently found in old volumes.

Towards the close of the last century the silhouette for a time was superseded by pastel portraits. However, it was not long before the art was again in favor, and practiced by numerous artists. The brush, which had heretofore been almost exclusively used, was now supplemented by the knife or scissors, resulting in the art of "Psaligraphy," in which the portrait was either cut out of glossy black paper and then pasted on a white card, or the reverse, where the outline was cut with a knife from a sheet of white paper and then backed with a piece of silk, thus showing a black profile on a white ground. Specimens of the latter process are very rare, as it required an artist of no mean order for their execution.

Owing to the popularity of the silhouette it soon became elevated to the rank of meritorious art. Noteworthy among the exponents of this school we will mention the late Paul Konewka, Ströhl, and others, who produced effects of a surprising degree of naturalness in solid black. Half a century ago silhouettes of prominent persons, actors, danseuses, soldiers, orators, etc., were as common an article of commerce as the photograph of a professional beauty is at the present day.

The advent of photography eventually proved the death blow to the silhouette, as a picture with all the detail and expression was far preferable to the simple profile. Yet at first, in many cases, on account of poor posing or defective lighting, the photograph showed less similarity to the sitter than

the well-executed silhouette. At the present day, with the great advances in the photographic art, all necessity for the silhouette has ceased to exist, nor is there any special reason why that style of portraiture should be made by use of the camera, except as a pastime. Yet, strange to say, this subject has excited so much attention in Europe that it was made the special order of business at the April meeting of the Photographische Gesellschaft, in Vienna, where Herr Eisele, a prominent member, stated that he had experimented for the last two years in producing photographic silhouettes. Professor Luckhart and the artist Herr Schuer had also devoted much time to the subject without, however, succeeding as well as the first speaker.

The details of Herr Eisele's experiments certainly make interesting reading for us on this side of the water. He states that at first he covered a frame with tracing paper, then placed his principal between the sun and the screen, thus throwing a shadow on the paper. The camera was set up on the other side, so as to photograph this shadow, the lens, of course, pointed direct to the sun. He then made the attempt to shade the lens with an umbrella. He neglects to state how often he got the outline of the parapluie on his plate. Then he tried Blitz-pulver--sitter, camera, and screen in same position. Result as might have been foretold--a miserable failure.

Next the screen was placed in a doorway with a bright light back of the screen. The subject was placed in front of the screen, the room darkened, the camera being placed in front of the sitter and screen. A Blitz-lamp exposure of five to six seconds was then made. Result--not a single specimen that equals our old silhouette. So much for our Viennese photographic cousins. In these great United States we simply tack a piece of muslin in front of a window, place the subject directly in front, shut off all other light in the room, focus, fire, and develop. Result--a good sharp profile almost all the time.

JULIUS F. SACHSE.

SOME FURTHER DETAILS ABOUT PRINTS.

The drying of a photographic print after the final washings have been completed is a simple enough matter, and yet it is possible for the most exasperating failures to occur at this stage of the process; the disappointment experienced being all the more keen because the work is in a certain sense finished.

Those unacquainted with photographic neatnesses might easily imagine that all that was necessary was to take the print out of the water, and lay it aside in any convenient place to dry. They would soon find out, however, that if the substance with which the wet print came into contact were capable of communicating any impurity, the print would be sure to show in the form of stains. For instance, suppose that the prints were hung over wooden poles, or laid on wooden shelves while still wet, there would hardly be a possibility of escape from stains. This would be true in the case either of silver prints of any kind, bromides, or blue prints.

Silver prints on plain paper and blue prints are more manageable in drying than the other forms which are made on papers prepared with a contractile substance like gelatine or albumen. Supposing that the wooden poles or shelves before spoken of were covered with clean white linen or blotting paper, all those forms of prints having a plain surface might safely be dried there, but an albumen paper print would not do so well; if laid out flat on the shelf it would contract unequally, and be so crinkled and shrunk that there would be serious difficulty in trimming. Drying over the pole would be preferable, but the albumenized surface would be put on the stretch unequally, so that in the case of a highly glazed surface there would be fissures and cracks very detrimental to the finished result.

The best method of drying prints of all descriptions, and a very convenient and inexpensive one also, is the following: Provide a number of

spring clothes-pins, a few yards of ordinary brass wire, and a couple of good-sized screw-eyes. Having selected a suitable place in the workroom where the prints will not be disturbed, screw in one of the screw-eyes to the wood of the window or door jamb at the height of the shoulder; pass one end of the brass wire into the eye and secure it, then string the clothes-pins on to the wire, and secure its other end by means of the other screw-eye at a convenient point across the room. Having brought the prints from the washing tank in an ordinary deep pan, select those of similar sizes, bring them together neatly, back to back, while in the water, then take them out and suspend the pair from one or more of the clothes-pins, according to the size of the prints. If they are very large, it may require three of the clothes-pins to fully support them, and avoid risk of the wet mass tearing by its own weight. While on the other hand, small sizes, such as 5×4 inches, may be held by a single pin. When the paper is very glossy, and the weather dry, the larger sizes may require to be pinched together at the bottom corners by an additional couple of clothes-pins, which will prevent the prints from separating until thoroughly dry.

Prints dried in the manner described will be quite flat, and free from stains of any kind. We need hardly add that the clothes-pins should be new and clean, and kept for this purpose only. If the prints are hung up to dry in the evening, they will be ready for trimming in the morning, when the end of the wire may be released, and the whole turned aside out of the way until the next occasion for use. If the wash-water is muddy, as is often the case, the deep pan in which the prints are transferred to the drying-room may be filled with clean filtered water, so that the collected mud in the paper may be soaked out before drying.

The warm weather we are now passing through reminds us of a few matters which have greatly eased our own labors in the printing-room, and simple as they are we will mention them.

It sometimes happens that there is trouble in securing pure whites in prints on albumen paper, an universal yellow stain covering everything. The best remedy for this is the use of alum in the printing bath, as originally suggested by the late Mr. Anthony, of New York. Care also must be taken that the paper is quite dry before being fumed. Operators are too apt to forget that as the thermometer rises, so does the amount of watery vapor in the air increase, and that sheets of paper will often dry more quickly on a

bright day in winter than on many hot days in summer. The way the paper feels to the hand is the best guide, and some little attention is required to be able to tell accurately.

The question whether the strength of the silver bath should be reduced or not during warm weather is open to some discussion. If the paper be of first-class quality, and the bath contain alum, as before alluded to, it would be possible to continue making good prints having pure whites with the bath at full strength, by which we mean fifty to seventy grains to the ounce. There is no question of the fact that the sensitiveness of the prepared paper increases when floated on a strong bath, and that the compound which is then formed between the albumen and the silver is more prone to decomposition. It will occasionally happen, if the prints come out yellow, metallic-looking, and covered with minute black specks, that weakening the silver bath down to the strength of forty-five or even forty grains will cure the trouble. A strength of forty grains, however, we should consider a low one, and only to be resorted to for unusually hot weather or for particular kinds of paper, such as very thin and delicate Rives.

The paper should not be left in the fuming-box for too long a time in hot weather. If things are properly arranged for the purpose, ten to twelve minutes ought to suffice for thorough fuming. It is important, of course, that good strong ammonia be employed, and care should be taken that the glass stopper be well secured in the bottle. In a hot printing-room the stoppers of ammonia bottles are frequently blown out by the vapor and fall on the floor, leaving the contents of the bottle to lose strength rapidly.

ELLERSLIE WALLACE.

THE CONVENTION AT WASHINGTON.

The time for the Eleventh Annual Convention of the Photographic Association of America is approaching, and we hope the craft through the length and breadth of the land are preparing for it. Mr. J. M. Appleton, the president, sends the following circular, which should be carefully studied by every photographer, and also a specimen of the blank form to be filled up by intending exhibitors. Those who may not have received a copy should apply to him.

We published a list of the prizes to be awarded in our March number, and now print the following:

RULES FOR JUDGES AND EXHIBITORS:

RULES GOVERNING JUDGES IN THE GRAND PRIZE.

1. The points to be considered are: First, historic; second, originality; third, composition; fourth, lighting; fifth, technique.
2. Ten marks to be the highest for any one point, consequently, fifty marks the most that can be given to any one picture.
3. The standard of this award must be thirty-five markings out of a possible fifty.

RULES GOVERNING JUDGES IN CLASS A.

4. The principal points to be considered are: First, originality; second, composition; third, lighting; fourth, technique.

RULES GOVERNING JUDGES IN CLASSES B, C, D AND H.

5. The following points to be considered are: First, lighting; second, posing; third, chemical effect; fourth, general effects or finish.
6. Ten marks to be the highest for any one point, consequently forty marks the most that can be given to any one picture.
7. The above regulations also to govern Class A.
8. Applications for space must be made to George H. Hastings, No. 147 Tremont street, Boston, Massachusetts.
9. Entries to close on Tuesday, July 15, and no space to be allotted for exhibits after that date. All exhibits must be shipped so as to reach the exhibition on the Saturday preceding the opening of the same, and all charges to be prepaid.
10. Each picture or set of pictures must be marked with a letter signifying the class in which it is offered for competition.

11. The exhibition of photographs connected with the convention is to be considered an art exhibition pure and simple. In order not to detract from this standard, no sign of any description shall be allowed in the hall devoted to the display of photographs, except the names and addresses of the exhibitors.

12. No exhibit for the grand prize or classes A, B, C, D and H to occupy space of more than eight lineal feet, and in class E, ten lineal feet.

13. All photographs exhibited must be from negatives made since the tenth annual convention, held at Boston, Massachusetts, August, 1889.

14. All exhibitors must see that their displays are properly placed before 10 o'clock A.M., on August 12, and no exhibits will be admitted after that date.

15. All art exhibits must be sent to George H. Hastings, Art Department, Photographers' Association of America, Smithsonian Institute, Washington, D. C., all charges prepaid.

**ELEVENTH ANNUAL CONVENTION OF THE
PHOTOGRAPHERS' ASSOCIATION OF AMERICA.--
SMITHSONIAN INSTITUTION, AUGUST 12 TO 15.**

OFFICE OF THE SECRETARY, }
IOWA CITY, Iowa, May 24, 1890. }

To the Photographers of America:

The eleventh annual convention of the Photographers' Association of America will be held in the Smithsonian Institution at Washington,. D. C, August 12 to 15, inclusive, and from present indications the attendance will be larger than ever before. Why not take a few days' vacation and attend this meeting? If you only knew the treat that awaits you in Washington, every photographer in the land would attend, if they had to close their business for a week to do so.

A visit to the places of interest in Washington, the most beautiful city in the world, will alone be worth the time and money spent. The fine collection of paintings in the Corcoran Art Gallery is of special interest to photographers, as it contains some of the finest pictures in this country or in Europe. A visit to the capitol and government buildings, the Washington monument, the White House, the Smithsonian Institution, and many other interesting points will add to the pleasure of this occasion, and is alone well worth the time and expense of the trip.

The Committee on Railroads have secured a rate of one and one-third fare for the round trip on all trunk lines in every direction. In order to obtain this rate, you must obtain a receipt from the ticket agents of whom you purchase tickets on all roads, and have them signed by W. V. Ranger, second vice-president, at the convention, this will entitle you to a one-third fare returning.

The headquarters of the convention will be at the Ebbitt House, and the following rates have been obtained at the different hotels: Ebbitt House, two in a room, \$2.50 per day; Ebbitt House, one in a room, \$3 per day; the Arlington, \$3 per day; Millards, \$3 per day; the National, \$2.50 per day; the

Riggs, \$3 per day; the Harris, \$2 per day; and many other hotels from \$1.50 to \$2.50 per day; also several on the European plan, rooms from 50 cents to \$1.50 per day.

If you will look at the benefits to be derived from attending this meeting, we feel sure that you will take the time and meet with us.

The unveiling of the Daguerre memorial during the convention, to be permanently placed in the Smithsonian Institution, will be one of the special features. This memorial is the gift of the photographers of America. The fund is raised by one or more dollar subscriptions from the photographers (which it would be well for you to send in your subscription at once to one of the different committees, and help the matter along, and do honor to the man who first brought to light this noble profession of ours).

You will miss it if you *fail to come*. There will be some of the finest specimens of American photography ever exhibited, as well as European. I have the promises of exhibits from all the leading photographers of the world.

Rules and regulations, list of awards, entries for competition, etc., will be mailed on application.

Is not this association worthy of your support? Any photographer of good moral and professional standing is eligible. If not already a member, lose no time in uniting yourself with an organization already a power in the land. It has stood the test of time (eleven years), and has a creditable standing throughout the length and breadth of the civilized world.

To become a member, send five dollars if a proprietor, and two dollars if an employé (which pays entrance fee and dues for first year), to the treasurer, G. M. Carlisle, Providence, R. I. If already a member kindly remit your dues (\$2.00), and by so doing avoid waiting your turn at the entrance when you arrive at the convention, as none can be admitted whose annual dues remain unpaid.

EXTRACT FROM CONSTITUTION.

ARTICLE II., Section 4.--The annual dues become payable on January 1 of each year, and any member failing to pay the same prior to the adjournment of the annual convention shall forfeit his right to membership, and can only be reinstated on payment of an initiation fee (\$3.00) and (\$2.00) dues, \$5.00, as provided in case of admission of new members.

Become a member and get the benefits of the art lectures, practical talks, and a grand exhibition of photographic productions and stock exhibit.
Fraternally yours,

D. R. COOVER, *Secretary.*

[We expect to publish, in our next number, an additional article on this subject, from the pen of the president, J. M. Appleton, which will reach our subscribers in ample time for the convention.]

S.

The average newly-fledged photographer wishes to begin before breakfast. The barn or the back-yard fence is good enough for him, and if nothing else alive is in sight, it is easy enough to try it on the dog. Few dogs, however, have a proper sense of the responsibility of being photographed. After everything is carefully arranged, they get up and want to go home. Sometimes they remain silent until the picture is started, when they begin to stroll around and make eight or nine dogs on the plate; or they sit still and only wag their ears, until the negative shows something like a halo of ears. This would give a professional photographer much worriment of mind, but the amateur is generally wise enough to see that his failures are the funniest things he has.--*Photo. Times.*

ORTHOSCOPIC PHOTOGRAPHY.

It is well-known that there are a large number of coal-tar color products, which have the property of causing an orthoscopic action on the ordinary sensitive dry plate, making the plate more sensitive for a certain color than the others; for instance, eosine is a superior sensibilator for green-yellow and yellow-green; cyanin, again, is especially sensitive for reds. Other products, such as rhodamin, cyclamin, methyl, violet, and others, generate in each instance an especial color sensitiveness.

While we have no deficiency in these mediums which answered for the greens and yellows, we have but few that answer for the various shades of red. Although Dr. Vogel has strongly recommended a mixture of chinolin red and chinolin blue (cyanin), the latter has two great drawbacks, viz., the salt is very dear, and does not keep. To overcome this difficulty I have instituted numerous experiments to replace the above with a more durable and cheaper medium. Among the various substances which I have tried, the mixture of indophenol and malachite green, has given me the most satisfactory results. Malachite green alone produces a strong sensibility to red, but the addition of the indophenol greatly enhances this sensibility.

In connection with the experiments with this solution, the interesting fact was observed that the sensibility for blue was greatly reduced in the brom-gelatine film, while green and yellow appeared in their true color values.

My proceeding is as follows:

(A.) Dissolve 0.1 g. indophenol (naphthalin blue) in 500 ccm. alcohol.

(B.) Dissolve 1 g. malachite green in 200 ccm. water.

The latter solution is heated to about 70° c.

In the meantime, prepare a solution of 10 g. doppelt chrom sauris kadi in 100 ccm. water, at a temperature of 70 to 80°, then pour this to the hot

solution of malachite green.

This mixture is kept hot for half an hour and then filtered, the precipitate which remains on the filter is now washed in several waters, and finally again dissolved in a solution of

Alcohol	250 ccm.
Chinin sulphate	0.8 g.

The latter is first dissolved in a little alcohol by heating, then the volume is added to until the amount is reached. The filtered fluid has a beautiful greenish-blue color. This forms the stock solution.

To sensitize the plates, make the following bath, pour in graduate:

Indophenal solution (as above)	4 ccm.
Malachite green (stock solution)	4 ccm.
Water (distilled)	600 ccm.

Pour 60 ccm. of this solution in a tray (13 by 18 cm.) cover the plates and soak 2 minutes, keeping the tray in constant motion. During the operation all red rays must be carefully excluded; then the plate is drained and dried in absolute darkness.

Plates so treated are much quicker than when in their normal condition. They give the reds in their true color value, even through an intensive yellow color screen. Even this extreme color sensibility can be enhanced with the use of a "supplementary-ray filter" ("Ergänzungs-Strahlen filter"), which is made from a substance which absorbs all colors except the reds and yellow. Gelatine, dyed by soaking in a erythrosin solution, furnishes an excellent Ergänzungs-Strahlen filter, which is used in connection with the usual yellow color screen

M. LEON VIDAL.

PHOTOGRAPHY IN ROME.

The *Bollettino*, a journal published every two months in Rome, and the chief organ of those interested in photography in that city, gives the result of the last election of members of its Photographic Society as follows: Adler, Dottor Vittoria; Intriglio, Avv. Benedetto; Tenerani, Cav. Carlo; and of the correspondent members--Calvaria, Cav. Avv. Giuseppe, of Castellamare di Stabia; Daniele, Oreste, of Catania; Garzia, Oronzo, of Maglie; Hermans, Charles, of Brussels; and Orsini, Marchese Antonio, of Solmona.

The opening of the Second Annual Exhibition of Photography took place in the Palazzo di Belle Arti on the 2d of May, when a large number of members took part.

The display included landscapes and interiors, portraits, instantaneous views, film negatives, photo-micrographs, enlargements, flash-light pictures, architectural views, and representations of costumes taken in Morocco and in Sweden.

An excellent article on "The Duration of the Pose" begins thus:-
-“Formerly, when collodion occupied the whole field of photography, the first difficulty in the art was the preparation of the sensitive surface. Now that the preparation has undergone a radical change, passing into a branch of industry, that first difficulty has vanished, and there has succeeded to its place the duration of the pose, the exact determination of which has over the resulting picture an influence as great, and possibly greater than formerly, seeing that in the very short poses of the present time even a little error has a value of relatively greater importance.” The article goes on to say that the duration depends (1) on the sensitive preparation, (2) on the actinic power of the light, (3) on the object, (4) on the diaphragm, and (5)

on the distance. Thus, if it were required to represent by an algebraic formula the conditions governing the pose t'' , we should have--

$$t'' = k. P. L. O. d. D.$$

where k is the invariable constant.

The article, which is too long to translate, is written by A. Roncalli. It is succeeded by a short notice of the effects of the Schippang varnish upon collodion enlargements. This article, written by Ab. F. Castracane, makes mention of some unhappy results of the use of this varnish on some of his own pictures. After this comes a letter from Sac. D. Ratti, on halation, or *aureole*, as the Italians call it. Then a paper on the development of instantaneous negatives and on the toning of aristotype paper, by Bne. T. Melazzo. Various notes and receipts, with a bibliography and a short notice of the illustrated supplement, complete this interesting number. The illustration is that of a moonlight effect, the negative by A. Ducros, the phototype by Danesi, of Rome. "To obtain this picture," says the letter-press note, "it was sufficient to set the machine against the setting"--pardon the indiscretion; I was about to add sun--"and to remove the cap. But, before this is done, that certain fifth sense has to be taken into account, without which,

‘Non licet adire Corintum!’

and Signor Ducros, profound and advanced artist and photographer as he is, possesses this fifth sense, and uses it in a masterly manner.”--*Photographic (London) News*.

AMATEUR EXPERIENCES.--IV.

If any one wants to become thoroughly acquainted with the weaknesses and frailties of humanity, just let him become a camera carrier, in "all that the word implies,"--and he will enter a school, wherein he will learn more of the different phases of human nature in one lesson than he has during the last ten years of his life. No other vocation, if we except that of the live newspaper reporter, offers the same advantages in this biological study. Varied indeed are the experiences and vicissitudes of the amateur photographer, whether the camera bearer carries the latest Universal, with aluminium mounting, or rejoices in a Premium Pinhole outfit, he experiences the same annoyances and disappointments. Ignorant and unreasonable people are sure to be met with on an outing, and, worse than all, he has frequently to suffer for the sins of some rude member of the guild who has been there before him. Experiences like these are but too apt to discourage persons of a nervous or sensitive temperament; the picture, however, is not all shadows. There is often a bright side for the camera bearer, especially if he be susceptible of the humorous. Photographically speaking, the writer, in addition to such annoyances as double exposures, unaccountable fog, forgetting to draw the slide, put plates in the holder, or take the cap off, to say nothing of neglecting to insert the stop, has met with many rebuffs and disappointments on his outings, through meeting with ignorant or unreasonable people, in all such cases his rule has been always to look upon the comical side of the situation, and try to achieve his object, bringing into play his common sense, tact and knowledge of human nature, generally with the result of obtaining the coveted negative.

The trouble, however, does not always lay in the strangers we meet in our travels; the fault too often is with the camera bearer. There is a class of persons, largely represented among the guild of amateur photographers,

who presume entirely too much on their wealth or social standing, and who at home pride themselves on their good breeding and polite manners, claiming to be within the so-called exclusive social circles or sets; yet they no sooner get away from the restraint of their immediate surroundings, such as a photographic outing affords, than they seem to forget that at least a little courtesy is due the strangers on whose premises they trespass. The dweller in a picturesque tumble-down shanty, or custodian of an old colonial or religious land mark, no matter in how humble circumstances of life they may be, have rights guaranteed them under the law, which even the exclusive amateur is bound to respect. One specimen of this kind will often spoil the game for all amateurs for some time to come. We will give a few instances which have come under our notice.

Early last spring two prominent members of the Quaker City Camera Club concluded to photograph an historic old church in one of the German counties (so-called) in our state. The place was rather difficult of access, being away from the usual lines of modern travel, so extensive preparations were made. The day proved all that could be desired. A team had been telegraphed for, and met the pair at the nearest railroad station. When the spot was reached the outfits were quickly unpacked and set up. No permission was asked, nor notice was taken of flower-beds trampled over, or other damage done. Then the bell of the parsonage was rung,--the clergyman answered the call in person. The spokesman, great in his own importance, asked for the keys of the church, as they wished to photograph it. The dominie answered, with an unmistakable Pennsylvania Dutch accent, that the sexton lived about a half a mile down the road. The reply was, that as he had a key it did not suit them to run after the sexton. Well, one word brought on the other, and the parley ended with the clergyman saying, "You will please excuse me; I got no time to fool with such nonsense, and I can't be bothered with opening the church for every fool photographer who chooses to come out here from town; please go about your business, if you have any." The interview closed with the threat by the parson to use a hoe-handle over the next photographers who should come to bother him with their intrusiveness. Our two amateurs packed up their outfits and beat an ignominious retreat, going back to Philadelphia with temper ruffled and object unaccomplished.

A few days after this episode one of the twain, wishing to have some sport at our expense, suggested to the secretary of the Leopardville Camera

Club the advisability of a trip to the adjoining county, and securing for him a set of negatives of the old landmark. We knew nothing of what happened, and consequently, owing to our innocence and unsophisticated nature, we unconsciously fell into the snare that our kind friend laid for us on the first fine day. We made an early start. We went merrily over hill and dale, not dreaming of trouble. When within half a mile of the church, we stopped at a roadside inn to water our horse and inquire our way. We remembered the hostler, who, while reigning up, caught a glance at our outfit. To our surprise, he broke out with, "Say, Mishter, vas you a fotegraf feller?" "Why?" we queried. "Vell, if you vas, tont go to der kärch up dere; der breecher letzt woch putty near broke zwei fotegraf feller's het's up." Here was a revelation. Our nickel had been well invested. We made closer inquiries,--from the description, we at once recognized our friend from the Quaker City who had suggested our trip, and was no doubt chuckling in anticipation of a countryman's discomfiture. After a few moments' thought we continued on our way. We met the dominie, and when we got back to our home we had eight negatives, exteriors and interiors, in our satchel. In one of the latter the dominie appears in the quaint old pulpit. With our ten dollar outfit, by the use of civility, tact, and common sense, we had accomplished that in which our predecessors had so signally failed, mainly by not exercising the common civility due towards a stranger.

Another case which came under our notice but a few weeks ago: In an adjoining county there still exist several quaint old buildings, erected during the middle of last century by a religious community, and used by them until long after the Revolution. Owing to the curious architecture and proximity to a summer resort, the property is often overrun by visitors and sightseers, who run over the grounds, enter the houses, intrude on the privacy of the inmates, as if they had no rights of their own whatever, and in fact act as if the whole premises were public property. The custodian or trustee of the property is a plain country Dutchman, and keeps an especially sharp lookout for amateur photographers, as the religious sect to which he belongs frowns down portraiture of any kind. A few weeks ago a party of nine or ten persons, ladies and gentlemen, made a pilgrimage to the old settlement under the leadership of a well-known pulpit orator in Pennsylvania. Among the party were several amateur photographers. When the party arrived at the grounds they entered, without as much as asking permission, and at once made themselves at home within the premises

regardless of the inmates. While the amateurs were setting up their cameras, the preacher was airing his knowledge of the religious doctrines of the community which flourished there in days gone by. While making these derogatory allusions, the party had been joined by a stranger,--it was the trustee, and who lost no time in introducing himself to the preacher. The two men were a study for an artist. The preacher, who prided himself on his fine physique, oratorical powers, and dignity, was the ideal picture of the petted fashionable preacher of the present day. The other, a man of medium height, bare-footed, unkempt; a straw-hat of last season's growth, a shirt of unbleached muslin, a pair of overalls, which hung by a single "gallus," completed his wardrobe; his language was pure and unalloyed Pennsylvania Dutch.

In appearance the two men as they faced each other were as diametrically opposite as the poles. The trustee, without any ceremony, asked the preacher what he was doing there; the latter, looking down at the speaker with contempt and scorn, and nettled at the interruption, curtly told him to attend to his own affairs. This was more than the trustee could stand, and he at once ordered the party to pack up and get out. This in turn was too much for the preacher; so, turning to the trustee, said, "My good fellow, you seem not to know whom you are addressing; I am the Rev. Dr. ----, of ---- Church, in Philadelphia, and I wish you to understand that you are in the presence of ladies and gentlemen, and I would advise you to take yourself off without ado, as your presence here is unwarranted, uncalled for, and distasteful to the persons present as well as myself personally; and further, your appearance is hardly such as would be permitted within the circles in which the ladies present are in the habit of moving." During this speech the trustee stood with mouth and eyes wide open. The others of the party nodded approval as their spiritual leader was delivering himself. One of the photographers was trying to train his camera on the countryman, who had for a few moments stood speechless. But it was only the calm before the coming storm. With a bound the trustee kicked over the tripod and camera; then, turning to the preacher in an unmistakable attitude, told him in his rich German English, that he was on private property, tramping down a growing grass crop, and if he and his crowd didn't pack themselves off at once he would arrest and fine the party for trespassing. "But, my good fellow," ventured the now crestfallen preacher. "Don't speak to me!" was the retort. "You claim to be a gentleman; maybe you try to be at home. But if you

were one, you would know better than coming with a crowd on another's place, where you have no business, without even asking permission." "But, my good man, we are willing to pay you if--" broke in the preacher. "We don't want your money. All I want is for you to go and not bother us. Or do you want me to show you the way?" All this was said in the rich vernacular peculiar to the locality. There was no help; the trustee was on his own ground. So the party retreated and filed singly over the old stile into the road. It would be hard to say which of the party felt the sadder as they wended their way towards their conveyance, the crestfallen preacher or the Rittenhouse amateur with his shattered outfit.

This was but another instance where a little courtesy and politeness would have saved humiliation and photographic disappointment.

J. FOCUS SNAPSCHOTTE.

A RECORD IN DEVELOPMENT.--Many amateurs are so fidgety about their dark-room and its appendages that we describe, both for the benefit of the finic and also for "those who go down to the sea"--in trains, how an extemporized traveling dark-room was successfully used by a member of the newly founded Croydon Camera Club.

In case the railway superintendent should reprimand the guard who connived at the measures adopted, we must perforce suppress the gentleman's name, the date, and the station where the train was joined.

About a fortnight ago Mr. X. ran down for the day to visit his friend Y., who dwells somewhere on the south coast, within about one and a half hour's journey from Croydon. Mr. X. having exposed sixteen quarter-plates, Y. enquired of him when they would be developed. "To-night," answered X., and added, "Perhaps before I get to Croydon." Y. expressed incredulity, on which X. guaranteed that he would have all the plates developed before reaching his destination.

No previous preparation had been made, and the train started in forty minutes from the time of above conversation. A sheet of ruby paper, some drawing pins, some oiled paper, and a piece of Willesden waterproof paper, together with Beach's developer, in two solutions, were procured. The guard was duly "tipped," and a pail of water obtained from the engine-driver. Mr. X. being safely locked in a third-class compartment, the

Willesden paper was made into a tray, with sides three inches deep, on account of the swaying of the train. The ruby paper was pinned over the carriage lamp, and the blinds carefully drawn. The night was, fortunately, a dark one. Most of the plates were shutter views; these were first developed, the developer being used for about three plates and then thrown away. The time views were subsequently developed, with a suitable modification in the proportions of developer. The plates were well rinsed in the pail of water, and while wet wrapped in oiled paper, and thus packed in the ordinary boxes in which they are sold; the object of using oiled paper being that it does not stick to the film when the latter is either dry or wet. The plates were all developed before Red Hill was reached; the fixing being deferred until arriving home.

The resulting negatives were not noticeably inferior to those which the same worker generally produces in his dark-room. We have before us a print of a wreck with fisher-boats “salving,” which is distinctly above the average skilled amateur work.

If so good a result is attained by adapting a railway carriage on the spur of the moment, even better could be done by pre-arranging to make use of the dreary time spent in traveling by night. The above *tour de force* is a strong argument in favor of those railway companies who run journeys of from five hours upwards, such as the Scotch express, providing a well-fitted but inexpensive dark-room. A luggage van might be converted, with an open compartment for workers to sit in when their “dark deeds” are done.

A pleasant vision is opened up of snap-shot views, taken from a railway carriage, and developed during the journey. Of course, plates need not be exposed while the train is “hurtling” along at seventy miles an hour; but in a, say, ten hours’ journey there are many stoppages and slackenings of speed which a member of the “wideawakes” could profitably utilize.--*The Amateur Photographer*.

MILITARY PHOTOGRAPHY.

Captain Curties, of the Royal Engineers, has written a series of articles on the above subject, which were published in the *Broad Arrow*. In the last of the series he gives a description of his photographic outfit. His arrangements for exposing and developing the plates in the field afford interesting reading. However, judging from our own experiences on the scout, the picket line, or field, we should say that the whole scheme, as portrayed by Captain Curties, is more or less chimerical, and no matter how plausible the plan may seem or read to the members of a theoretical camera club in their well-furnished quarters, there are certain difficulties in the way which would make the scheme impracticable, and even if these were overcome, the results would be of but little if any use in actual service, a fact which will be apparent to anyone who has seen active military service.

The captain in his articles says:

“The one object I have kept in view all through has been to simplify the art as much as possible, and to make each photographic section complete in itself, and able to take, develop, and print a picture without any outside help in the shortest time possible. Having this end in view, my equipment supplies in the first place two light knapsacks, to be carried in a reconnaissance by two mounted officers or men. One contains a very light fold-up camera, capable of taking pictures 10 in. by 12 in., round which is wrapped the focussing cloth. It has not appeared to me desirable to place before a general a view of a country smaller than this. The extra weight would be more than compensated for by the comprehensive picture obtained; moreover, in a small plate, I take it, distance would not be fairly and distinctly portrayed. The other knapsack carries three dark slides, very light but strong, each containing two plates. This knapsack also contains the lens, instantaneous shutter, etc. Both knapsacks are made to fit close to the

back, and, in addition to the straps passing over the shoulders, are secured to the sword-belt, thus preventing any injurious motion when riding. By the simple act of unbuckling one strap, each can be at once unslung ready to be unpacked. The tripod, which is made as light as can be consistently with strength and stability, is carried folded up in a bucket attached to the saddle, and fixes immediately on to the camera. We next proceed to the all-important subject of “developing” in the field. For this purpose I use a tent composed of a large, folding, umbrella-shaped top, made of a material which admits a deep ruby light. When this is opened and fixed in the ground, it stands just clear of a tall man’s head. Over it is dropped a sort of sack, open at bottom ends, the top end being much smaller than the bottom end, and capable of being drawn together by means of two cords. This sack is lightproof and waterproof. The lower end is held down by means of a light iron hoop or ring, which also folds up to facilitate packing. The hoop is attached to the bottom of the sack in such a way that a border of the material extends beyond it, and rests upon the ground. This, in the ordinary way, is sufficient to keep out light, but should any find its way in, a few handfuls of earth or stones heaped up round the border will effectually keep it out. The stick of the umbrella is a hollow bamboo, open at the top. It is pierced with holes to about half its length; this ventilates the tent. A cap placed over the top of the stick excludes light, but not air. We now have a complete tent in which a man can move about freely, and use his hands without constraint, and, above all, he is not half stifled, as one generally is in the usual run of developing tents. It can be taken down at a moment’s notice, and packed in a very small compass, the whole being exceedingly light and compact. A few stays may be necessary in windy weather to keep it steady. The person about to develop a plate slings over his shoulders, knapsack fashion, a small metal tank, containing sufficient water to wash several plates; attached to it is a gutta-percha tube and tap. Round the waist is buckled a broad leather belt, in which are fixed bottles containing the developing solutions, etc. A light fold-up trough, with a gutta-percha drain-pipe, carried outside the tent, and two light shelves, hook on to the stick of the umbrella. All that now has to be done is to lift up the walls of the tent, step inside, and develop and print the picture, which by using bromide-paper (undoubtedly *the* process for military use), would take something like a quarter of an hour, the printing, of course, to be done from the wet plate. I may mention that I use scarcely any glass beyond the plates (which I

believe in); those articles which are made of glass are protected to prevent breakage. I believe myself that the whole of the articles now made of glass can be manufactured from a preparation of celluloid, which is strong, light, and durable. I hope shortly to have a complete set of bottles, measures etc., made of celluloid.”

M. STRAVOS ZELLIS, of Alexandria in Egypt, recommends the following process for marking or lettering on the sensitized paper such names as we wish to give the prints. He takes a piece of thin white paper and traces upon it the words which he wishes to have at the bottom of his negative, and oils it on both sides; having removed the excess of oil by rubbing it between two sheets of bibulous paper, he coats it with varnish on both sides and allows it to dry. On the other hand, he removes from the bottom of the negative a portion of the gelatine, equal to the size of the paper, and substitutes for it the paper, which he sticks by means of a solution of gum arabic and water. He removes then the air bubbles, which would prevent complete adherence, and this being done, waits for his work to dry. If, when printing on the sensitized paper, it is found that the letters do not show very white, the defective portions should be retouched on the back of the oiled paper. To write his name Mr. Zellis makes use of a mixture of gum arabic, lampblack, and water. This process is simple, cheap, and gives excellent results.--*Annals Photographique*.

At the meeting of the Photographic Society of Berlin, President Stolze exhibited the sketch of the Daguerre monument to be erected in Washington city, at the cost of six thousand dollars.

Dr. Julius Stinde declared never to have seen anything more disgusting (schauerlicheres) than this unhappy head of Daguerre, crushed under the weight of a large ball, and attributes the depravity of our taste to the high duty on articles of fine arts. He says that such monstrosities show that we Americans are yet in point of art barbarians of the purest water. Mr. E. Himly, as well as Dr. Stolze, takes our part, and shows that the American photographic journals have unanimously condemned and ridiculed Mr. H. McMichel's scheme, and exonerate us as a body. Bravo!--*Photographische Nachrichten*.

DAGUERRE.

On the twelfth of August, in front of the Smithsonian Institute, in Washington, dedicated to manifold arts and sciences, will be erected a lasting memorial to Daguerre, the author that we all know fixed the visible image on a given surface, which is photography, with all its varieties and names, and they are numerous.

Why should we Americans put up such a memorial? The inscription on the granite below the bronze portrait tells the story:



“To commemorate the first half century in photography, 1839–1889. Photography, the electric telegraph, and the steam engine are the three great discoveries of the age. No five centuries in human progress can show such strides as these. Erected by THE PHOTOGRAPHERS’ ASSOCIATION OF AMERICA, August, 1890.”

The monument, now almost complete, its bronze features being of a high order of art, will stand sixteen feet high, and will be the only international monument in the city of Washington, where Smithson himself dedicated his fortune for the advancement of science in the western world.

In connection with this celebration we present to our readers a portrait of Daguerre. Strange to say, there are but few authentic portraits of Daguerre in existence. In our search for such a portrait we discovered the one here reproduced; it was engraved by Orr from a photograph by Dr. Meade, of New York, for the International Magazine, of that city, and used to illustrate

the obituary of the eminent Frenchman, September 1, 1851. As everything relating to Daguerre cannot but prove of interest at the present time, we republish the interesting article in full as it appeared at the time:

Lewis Jacques Mande Daguerre, whose name is forever associated with the photographic process, of which he was the discoverer, died on the tenth of July, in Paris, in the sixty-second year of his age. He was a man of extreme modesty and great personal worth, and was devoted to art.

He was favorably known to the world before the announcement of his discovery of the Daguerreotype. His attempts to improve panoramic painting, and the production of dioramic effects, were crowned with the most eminent success. Among his pictures, which attracted much attention at the time of their exhibition, were, "The Midnight Mass," "Land-slip in the Valley of Goldan," "The Temple of Solomon," and "The Cathedral of Sainte Marie de Montreal."

In these the alternate effects of night and day, and storm and sunshine, were beautifully produced. To these effects of light were added others, from the decomposition of form, by means of which, for example in "The Midnight Mass," figures appeared where the spectators had just beheld seats, altars, etc.; and again, as in "The Valley of Goldan," in which rocks tumbling from the mountains replaced the prospect of a smiling valley.

The methods adopted in these pictures were published at the same time with the process of Daguerreotype, by order of the French Government, who awarded an annual pension of ten thousand francs to Daguerre and M. Niepce, Jr., whose father had contributed towards the discovery of the Daguerreotype. Daguerre was led to experiments on chemical changes by solar radiations, with the hope of being able to apply the phenomena to the production of effects in his dioramic paintings. As the question of the part taken by him in the process to which he has given his name has been discussed sometimes to his advantage, it appears important that his position should be correctly determined. In 1802, Wedgwood, of Eturia, the celebrated potter, made the first recorded experiments in photography; and these, with some additional ones by Sir Humphrey Davy, were published in the journals of the royal institution. In 1814, Mr. Joseph Nicephore Niepce was engaged in experiments to determine the possibility of fixing the images obtained in the camera obscura; but there does not appear any evidence of publication of any kind previously in 1827, when Niepce was in

England. He there wrote several letters to Mr. Bauer, the microscopic observer, which are preserved and printed in Hunt's "Researches on Light."

He also sent specimens of results obtained to the Royal Society, and furnished some to the cabinets of the curious, a few of which are yet in existence. These were pictures on metallic plates covered with film of resin.

In 1824 Daguerre commenced his researches, starting at that point at which Wedgwood left the process. He soon abandoned the employment of the nitrate and chloride of silver, and proceeded with his inquiry, using plates of metal and glass to receive his sensitive coating. In 1829 Mr. Vincent Chevalier brought Niepce and Daguerre together, when they entered into partnership to prosecute the subject in common. For a long time they appear to have used the resinous surface only, when the contrast between the resin and the metal plates not being sufficiently great to give a good picture, endeavors were made to blacken that part of the plate from which the resin was removed in the process of heliography (sun drawing), as it was most happily called. Amongst other materials, iodine was employed; and Daguerre certainly was the first to notice the property possessed by the iodine coating of changing under the influence of the sun's rays. The following letter from Niepce to Daguerre is on this subject:

"81, LOUP DE VARVENNES, June 23, 1831.

"Sir and Dear Partner:--I had long expected to hear from you with too much impatience not to receive and read with the greatest pleasure your letters of the tenth and twenty-first of last May.

"I shall confine myself in this reply to yours of the twenty-first, because, having been engaged ever since it reached me in your experiments on iodine, I hasten to communicate to you the results which I have obtained. I had given my attention to similar researches previous to our connection, but without hope of success, from the impossibility, or nearly so, in my opinion, of fixing in any durable manner the image received on iodine, even supposing the difficulty surmounted of replacing the lights and shadows in their natural order. My results in this respect have been entirely similar to those which the oxide of silver gave me; and promptitude of operation was the sole advantage which these substances appeared to offer. Nevertheless, last year, after you left this, I subjected iodine to new trials, but by a different mode of application. I informed you of the results, and your answer, not at all encouraging, decided me to carry these experiments no farther. It appears that you have since viewed the question under a less desperate aspect, and I do not hesitate to reply to the appeal which you have made.

"J. N. NIEPCE."

From this and other letters it is evident that Niepce had used iodine, and abandoned it on account of the difficulty of reversing the lights and shadows. Daguerre employed it also, as it appears, with far more promise of

success than any obtained by M. Niepce. On the fifth of July, 1833, Niepce died; in 1837 Daguerre and Isidore Niepce, the son and heir of Nicephore Niepce, entered into a definite agreement, and in a letter written on the first of November, 1837, to Daguerre, Isidore Niepce says, "What a difference, also, between the method which you employ and the one by which I toil on! While I require almost a whole day to make one design, you ask only four minutes! What an enormous advantage! It is so great, indeed, that no person knowing both methods would employ the old one." From this time it is established that although both Niepce and Daguerre used iodine, the latter alone employed it with any degree of success, and the discovery of the use of mercurial vapor to produce the positive image clearly belongs to Daguerre. In January, 1839, Daguerreotype pictures were first shown to the scientific and artistic public of Paris.

The sensation they created was great, and the highest hopes of its utility were entertained. On the 15th of June M. Duchatel, Minister of the Interior, presented a bill to the Chamber of Deputies relative to the purchase of the process of M. Daguerre for fixing the images of the camera. A commission appointed by the Chamber, consisting of Arago, Etienne, Carl, Vatout, de Beaumont, Toursover, Delessert (Francois), Combarel de Leyral, and Vitet, made their report in July, and a special commission was appointed by the Chamber of Peers, composed of the following peers: Barons Athalin Besson, Gay Lussac, the Marquis de Laplace, Vicomte Simeon, Baron Thenard, and the Comte de Noe, who reported favorably on the 30th of July, 1839, and recommended unanimously that the "bill be adopted simply, and without alteration." On the 19th of August the secret was for the first time publicly announced in the institution by M. Arago, the English patent having been completed a few days before, in open defiance and contradiction of the statement of M. Duchatel to the Chamber of Deputies, who used these words:

"Unfortunately for the authors of this beautiful discovery, it is impossible for them to bring their labor into the market, and thus indemnify themselves for the sacrifices incurred by so many attempts so long fruitless. This invention does not admit of being secured by patent."

In conclusion, the Minister of the Interior said: "You will concur in a sentiment which has already awakened universal sympathy. You will never suffer us to leave to foreign nations the glory of endowing the world of

science and of art with one of the most wonderful discoveries that honor our native land.” Daguerre never did much towards the improvement of his process. The high degree of sensibility which has been attained has been due to the experiments of others.

Daguerre is said to have been always averse to sitting for his own picture, and there are but few photographs of him in existence. The one from which our engraving is copied was taken by Mr. Meade of this city, and first appeared in the *Daguerrean Journal*, a monthly periodical conducted by S. D. Humphrey and L. L. Hill, who were distinguished for their improvements upon Daguerre’s process.

GELATINOGRAPHY.

A very rapid process to make newspaper illustrations, called gelatinography, is described in the following:

A black glass plate or a tin plate coated with black varnish, as used by sign-painters, is covered with plaster of Paris (gypsum) to a thickness of four-ply cardboard. The plaster of Paris must be of the best quality and reduced to a very fine powder. Add thereto some alum and some sulphate of barium, and in order to prevent the coating from being too brittle, add also a trifle of glycerine or of a gelatine solution.

This mixture must have the consistency of a thin pulp when applied to the glass or tin with a soft camel's-hair brush.

When dry, the artist may engrave into this coat of plaster of Paris, by means of a lithographic engraving needle, any design or picture with the greatest ease; the plate or glass is thereby laid bare, and design or picture appears black through the plaster of Paris coating. Mistakes or errors are easily remedied by filling in the plaster of Paris preparation.

With the regular printers' roller composition a stereotype is now made of the picture or design on the glass or plate, in the usual way; some bichromate of ammonia solution should be added to the roller composition, to make the stereotype hard enough for the type press, and it will be as durable as any electrotpe, and answer the same purpose.--*Am. Lith. and Printer.*

STEREOSCOPIC PHOTOGRAPHY.--Of late, there is quite a revival in this branch of our art-science, several English and many foreign amateurs having been working with twin lenses during the last and present seasons. The Belgian *Bulletin* has an article on the subject, and the last technical

meeting of the Photographic Society was devoted to it. Although Wheatstone announced the instrument in 1838, it was not until photography had come to his aid by furnishing satisfactory diagrams, and Brewster had popularized the matter by the invention of the lenticular stereoscope, that much progress was made; then Wheatstone gave his Bakerian lecture on January 15th, 1852, to put the finishing touch to this important branch of scientific work. The earlier attempts failed by reason of employing too wide an angle.

ECLIPSE PHOTOGRAPHY.

Probably in no department of science, certainly in no branch of astronomical science, has photography been of such use as in the study of solar eclipses. It is only when the sun is obscured by the moon that we are able to see and properly photograph the corona or luminous atmosphere around the sun. This solar corona, as has been said by Young, "is visible only about eight days in the century in the aggregate, and then only over narrow strips of the earth's surface, and but from one to five minutes at a time by any one observer." Very little of the eight days, however, can be utilized; indeed, as has been pointed out by Miss Clerke in her admirable *History of Astronomy During the Nineteenth Century*, the corona has only been observed by scientific men during forty-five minutes in as many years. Opportunities of observing an eclipse occur therefore at such comparatively long intervals, the phenomena to be observed are so varied and extensive, and the time during which the observations must be made is so very limited, that any permanent records of the phenomena, such as photography enables us to obtain, cannot fail to be of the greatest value. The most careful drawings of the same eclipse by different observers at the same station are so very dissimilar that it is generally unsafe to base any conclusion on them; whereas in photographs we have truthful records of the actual phenomena without personal equation of any kind, and with the additional advantage that there is more detail in the photograph than it is possible to insert in any drawing made during an eclipse, or even at leisure after the three or four minutes' observation of such an indefinite and irregular object as the corona. The history of the increase of our knowledge of the corona is practically the history of the improvement of our photographic methods of attacking the phenomena of an eclipse.

The first occasion on which photography was used at an eclipse of the sun was on July 8, 1842, when Professor Majocchi, at Milan, attempted to obtain Daguerreotype pictures of the corona. His account of the attempt informs us that "a few minutes before and after totality an iodised plate was exposed in a camera to the light of the thin crescent, and a distinct image was obtained; but another plate exposed to the light of the corona for two minutes during totality did not show the slightest trace of photographic action. No photographic alteration was caused by the light of the corona condensed by a lens for two minutes, during totality, on a sheet of paper prepared with bromide of silver." No details are given of the apertures of the lenses employed, or of their focal lengths. At the outset, therefore, astronomers were met with failure, but the failure at Milan did not deter Dr. A. H. Busch and Herr Berkowski from a similar attempt at Königsberg on July 28, 1851. The telescope used on this occasion had an aperture of 2.4 inches, and a focal length of 30 inches. Commencing immediately after the beginning of totality, a plate was exposed for 84 seconds in the focus of the telescope, and on development an image of the corona was obtained. A second plate exposed for from 40 to 45 seconds was fogged by the sudden breaking out of the sunlight. The picture thus obtained--the first photograph of the corona and prominences--is known as the Königsberg Daguerreotype, and is still preserved at the Strasburg Observatory. It was lent by Professor Winnecke for the exhibition of scientific instruments at South Kensington in 1876. On it the prominences, and the lower portion of the corona extending about one-fourth of a solar diameter from the moon's limb, are distinctly shown, the encroaching of the prominences on the dark disc of the moon, owing to irradiation, being particularly evident.

Daguerreotype was again used for the annual eclipse of May 26, 1854, by Mr. Campbell and Professor Loomis at New York; by Dr. Bartlett and Victor Prevost, who obtained nineteen photographs, at West Point; and by Professor Stephen Alexander and Mr. E. H. Old at Ogdensburg.

Liais in 1858 obtained photographs of the partial phases, using wet plates. On one of these the moon can be seen projected on the corona before totality. With the introduction of the collodion process more sensitive plates were obtained, and a great advance was anticipated. At the total eclipse of 1860, July 18, Mr. Warren de la Rue, at Rivabellosa, in Spain, used wet plates. His instrument was one specially devised by himself for photographing the sun's disc for sun-spots, and is known as the Kew

heliograph. It is an ordinary equatorial mounting with driving clock, carrying a photographic object-glass, 3.4 inches clear aperture, and 50 inches focal length. The primary image is .466 of an inch in diameter, but before the image falls on the plate it is enlarged by an ordinary Huyghenian eyepiece to 3.8 inches diameter. The exposing apparatus for the ordinary sun photographs is an instantaneous shutter; this, of course, had to be abandoned for the eclipse photographs. Two plates were exposed during totality, the exposure being 60 seconds in each case, but only slight traces of the corona were obtained. At the same eclipse Father Secchi and Professor Monserat, working at Desierto de las Palmas, obtained good photographs of the corona, using an object-glass of .15 metre diameter, and 2.5 metres focus, the primary image being 23 millimetres in diameter. The plates were placed in the primary focus and according to Secchi, "all the phases of the phenomena are represented on the photographs." The original negatives obtained at Desierto de las Palmas of this eclipse have unfortunately been lost.

The next attempt at photographing the corona was on August 18, 1868, this being remarkable as the first attempt to use a reflector for the purpose. Colonel Tennant and Sergeant Phillips at Gunttoor used a 9-inch silver-on-glass mirror, by With, of 6 feet focal length, mounted equatorially by Browning on the Newtonian plan.

Unfortunately the weather was unfavorable, but plates were exposed through light clouds, the longest exposure being 10 seconds. The photographs obtained show the prominences sharply defined, but only slight traces of corona are visible. Mr. Sutton, at Mautawali Kiki, and Drs. G. Fritsch, H. Vogel, and W. Zener, at Aden, were, from atmospheric and other causes, unsuccessful with refractors.

At the eclipse of August 7, 1869, many attempts were made to photograph the corona. In all cases where the image was enlarged before it fell on the plate, slight traces of the corona were obtained; while Professor Winlock and Mr. J. A. Whipple, at Shelbyville, with a 5½-inch lens of 7½ feet focal length, obtained seven pictures taken in the primary focus, one with 40 seconds' exposure, showing more detail than had previously been photographed.

At this eclipse, Messrs. Hoover photographed the corona with a lens of 12 inches focus, and Professor Stephen Alexander also obtained

photographs at Ottumwa some of which give good ideas of the coronal structure.

At the 1870 eclipse, December 22, a 4-inch Dallmeyer lens (rapid rectilinear), stopped down to three inches aperture, and with a focal length of thirty inches, was used by Mr. Brothers at Syracuse. Wet plates were used, and the photographs were taken through light clouds, the best of the pictures having had eight seconds' exposure. Details in the corona are very well shown in these photographs. In discussing his results, Mr. Brothers says, "The photographs taken ... prove that the light of the corona is very actinic, and that several photographs of this beautiful phenomenon can be taken during the time of totality." He further adds, "That it is impossible to obtain satisfactory photographs of the corona either with reflecting or refracting telescopes as ordinarily used is, I think, now conclusively proved."

Professor Winlock, at Jerez, during the same eclipse, obtained two good photographs with ordinary telescopes; while Lord Lindsay, at Maria Louis Observatory, with a 12-inch mirror of 6 feet focus, obtained plates so much fogged as to be useless.

On December 21, 1871, splendid photographs were obtained at Baikul by Mr. Davis (Lord Lindsay's observer), and by Colonel Tennant, J. B. Hennessey, Esq., and Captain Waterhouse, at Dodabetta. In each case Dallmeyer 4-inch rapid rectilinear lenses of thirty-three inches focus were used, the exposures varying from five to forty seconds. Herr Dietsch, in Java, also obtained two good photographs with a "lens of short focus," with exposures of half and one-third second. Captain Hogg, at Jaffna, also got fair results with cameras 16 inches and 23 inches long. At the eclipse of April 6, 1875, Dr. Schuster, in Siam, obtained good photographs, although small, with an ordinary camera.

The eclipse of 1878 marked another departure in photography. Dr. Draper used wet plates, and got much detail in 165 seconds. Mr. Ranyard used Mawson & Swan's extra sensitive dry plates, with a 13-inch lens of 6 feet 2 inches focus, and obtained photographs extending 6' (one-fifth of a sun's diameter) from the limb with exposures of one and three seconds. Professor Harkness, the director of the American operations, arranged two cameras, with 6-inch Dallmeyer lenses of 37.9-inch focus, and Mr. J. A. Rogers and Mr. Clark with these, using specially prepared dry plates made by Mr.

Rogers, obtained two good series of photographs. In the report on the eclipse operations published from the United States Naval Observatory, Mr. J. A. Rogers not only discusses the value of photographs as compared with drawings, but enters fully into all the details of eclipse photography, concluding by strongly advocating the adoption of dry plates. Mr. O. L. Peers during this eclipse obtained a wet plate photograph showing greater extension of the corona than any of the dry plate ones, but there seems some doubt about the apparatus he used. He used either a 2½-inch or 3½-inch Voigtlander portrait lens, and exposed either for twelve or for twenty-three seconds. Mr. Peers says he used a 2½-inch lens, and twelve seconds' exposure, while Voigtlander declares he makes only 3½-inch lenses of the focus 1:8 Mr. Peers used, and on examination of the photograph it is found that the trail of the moon on the plate indicates an exposure of twenty-three seconds. After the 1878 eclipse dry plates were universally adopted by eclipse observers.

The photographic arrangements of the expedition to Sohag, in Egypt, for the eclipse on May 17, 1882, were made by Captain Abney, the chief objects of the expedition being to photograph the spectra of the corona and prominences. Arrangements were also made by Captain Abney for corona photographs with a 4-inch lens of sixty inches focus belonging to him. The spectrum photographs taken show as many as thirty lines in the prominences, while the photographs of the corona obtained by Dr. Schuster with exposures of from three to thirty-two seconds show great extension of the corona with the most exquisite detail. These plates are also remarkable for the discovery of a comet in the photographs, although the comet was not seen by observers. Captain Abney and Mr. J. Norman Lockyer were responsible for the methods of photographic attack adopted by the English observers, Messrs. Lawrence and Woods, at the Caroline Islands, on May 6, 1883. The spectroscopic results and the corona photographs taken with the 4-inch lens of Captain Abney, previously used in 1882, were most successful. Janssen on this occasion used two objectives, one 6-inch and one 8-inch diameter, and using long exposures, photographed the corona extending two diameters from the sun, this being much further than it could be traced with a telescope.

Photography was again used on September 8, 1885, at the total eclipse in New Zealand.

At the eclipse of August, 1886, visible at Granada, Captain Darwin used a chronograph as devised by Dr. Huggins, consisting of a mirror inclined in a tube in such a manner as to enable photographs to be taken in the primary focus without the intervention of a flat. Good results were obtained. Dr. Schuster and Mr. Maunder used 4-inch lenses of 60-inch focus, and obtained good results. Their spectrum photographs were also successful. Professor Pickering, of Harvard, used a heliostat and a photo-heliograph of 38 feet focus, supported horizontally, but no results were obtained with this apparatus, although he was partially successful with his other instruments.

Very few photographs were obtained of the eclipse of August 19, 1887, in Russia, owing to the unfavorable weather. The English observers intended to use similar instruments to those employed in 1886, but the weather did not permit.

The eclipse of January 1, 1889, was very successfully photographed by the American observers, the largest aperture used being thirteen inches. On some of the plates used during this eclipse the standard intensity scale recommended by Captain Abney several years ago was fixed, and for the first time definite conclusions as to the brightness of the corona were obtained.

The expedition sent out by the Royal Astronomical Society for the eclipse of December 22, 1889, were each fitted with a 4-inch photographic lens, belonging to Captain Abney, mounted on the usual equatorial plan, and intended to continue the series so well begun by Dr. Schuster in 1882, and also with a 20-inch mirror of 45 inches focus, specially constructed and mounted for eclipse work, and designed to photograph the outer portions of the corona too faint for ordinary instruments. The plates for use with the 4-inch lenses were specially prepared by Captain Abney, and on each of them he had placed a scale of standard intensity squares for measuring the brightness of the corona. Small squares on each of the plates were exposed to a standard light for various times; these squares were then covered with a strip of black paper, and the plates taken out to the Eclipse Station and exposed on the corona. When the plates were developed the image of the corona and the squares were, of course, developed to the same extent, the squares thus serving as standards for absolutely measuring the photographic intensity of the light of the corona. The density of the deposit in any part of the picture of the corona can be compared with the density of the most

similar of the squares on the same plate by Captain Abney's photometer, and as this photometer depends upon the method of limiting apertures, it gives absolute readings.

The African expedition was entirely unsuccessful, owing to clouds, but the expedition to Salut Isles, under charge of the late Father Perry, obtained successful photographs, which are at present under examination. From them Captain Abney will be able to measure the absolute photographic intensity of the light of the corona.

An American expedition was sent to Cayenne with instruments used on January 1, 1889, and obtained successful photographs, while an American expedition to Southwest Africa was unsuccessful, for the reason already given. This expedition, under the direction of Professor David P. Todd, was located at Cape Ledo, about half a mile from the English Eclipse Station.

Several new departures in eclipse photography were introduced. Chief amongst these was the remarkable apparatus by means of which no less than twenty-three objectives and two mirrors were accurately pointed at the sun and caused to follow it by one large clock. A large duplex polar axis (the old English form as used for the 12.5 inch reflector at Greenwich) was mounted on solidly constructed stone piers and very carefully adjusted. This axis is constructed of 6-in. wrought-iron tubing, the total weight being about 2000 lbs. In it the cameras were fixed by set screws, the optic axis of the instruments being adjusted parallel to each other, and at an angle equal to the south polar distance of the sun at the time of totality. The carefully regulated and very powerful clockwork attached to the instrument caused the polar axis to rotate, and thus the whole battery of instruments followed the sun. Each lens was fitted with a pneumatic shutter regulated to give the required exposure in each case. The cameras themselves were enclosed in a dark-room, the lenses only being exposed to the sun, so that dark slides were not required, the plates being held on open rotating frames, these frames being rotated at the proper time by pneumatic arrangements. When the cameras were once pointed, and the clock driving properly, all the operations of exposure and changing of plates were performed without personal superintendence by means of the pneumatic apparatus, and a chronograph attached to the valve system of this apparatus recorded the exact time at which each exposure was begun and ended.

It is to be regretted that this ingenious and elaborate apparatus did not have a satisfactory trial, owing to the dense clouds; but Professor Todd assures us that he was thoroughly satisfied with the success of the pneumatic movements during the three minutes ten seconds he brought it into operation at the time of totality.

It is not improbable that (in spite of the great strength and weight of the axis and the solidity of the supporting piers) with this plan of fixing a large number of cameras and spectroscopes on one polar axis, the constant opening and shutting of shutters, and the changing of the plates, may produce so much shake that none of the long exposure photographs will be satisfactory. This, of course, can only be ascertained by the use of the instrument on the corona, and several years must elapse before the trial can be made.

Another unusual instrument was a photo-heliograph of five inches aperture and forty feet focus, mounted on a combination of the equatorial stand and tripod.

The long tube was made of iron, coiled spirally and strongly riveted, the necessary rigidity being attained by strong wires extending from end to end, and tightly stretched by a disc in the middle of the tube. Close to one end of the tube the polar axis was attached by a universal joint; the other end of the tube being supported by two rods, one on the east and one on the west side, these rods being also attached by universal joints. By means of these rods the proper inclination was given to the tube. The east rod was the declination rod, and was capable of sliding along the polar axis. The west rod was for giving motion in right ascension, being terminated at the free end in the form of a piston of a sand clock fixed in an inclined position. The rate at which the sand escaped from the cylinder could be accurately regulated, so that the rate of descent of the piston was completely under control, and was, of course, such as would cause the instrument to follow the sun.

This instrument was erected at Cape Ledo, close to a hill of such inclination that the sun could be followed during the whole of the eclipse, while the long tube could be manipulated with greater advantage than would have been possible if the instrument had been erected on level ground. The hot air rising from the heated hill probably affected the

definition in the photographs, but under the circumstances that could scarcely be avoided.

This form of mounting certainly solved the question of the possibility of using long-focus lenses mounted as direct photo-heliographs, but the apparatus is certainly unwieldly, and was only got into the fit state that it was on the eclipse day by the very great care and patience of Professor Bigelow. As it was intended principally to photograph the partial phases of the eclipse with this instrument, instantaneous exposures were arranged for, but Professor Bigelow succeeded so well in the adjustment of the instrument and the regulation of the sand clock, that he would have tried to obtain photographs of the lower corona with it had the weather permitted.

The photographic apparatus on this instrument has a very ingeniously constructed revolving plate holder, carrying round plates of twenty-two inches diameter. The exposing apparatus and the apparatus for rotating the plate between the exposures were moved by pneumatic arrangements, exposures being made at intervals of six seconds, the exact time of each being recorded on a chronograph. As no dark slides were used, it was necessary to enclose the whole of the photographic apparatus in a dark-room. One hundred and ten exposures were made with this telescope during the partial phases of the eclipse, all the photographs taken having to be obtained through clouds.

In several expeditions previous to this, where more than one kind of observation has been required, two or more objectives have been mounted on the same stand and driven by the same clock; but this plan is always open to the objection that any accidental disturbance in the manipulation of one of the pieces of apparatus will most probably spoil the results for both. With the American plan of many objectives on one heavy axis, and a pneumatic apparatus to manage all the actual operations of exposures and changing of plates, this objection of possible accidental disturbance is to some extent overcome; but the shake of the many operations taking place on the one axis introduces another risk. Beside this, the apparatus is very heavy, and exceedingly difficult to transport and erect, even in a civilized country.

Such is a very summarized account of the instruments hitherto employed, and it seems to me that the time has now come when much can be gained by the employment of fixed instruments and a moving large plane mirror.

This idea of using a heliostat is, of course, not new, for it has been used several times on a small scale, and for special purposes. There is nothing beyond the difficulty of making a plane mirror sufficiently large for the work to prevent the adoption of this method in the future; and this difficulty now has ceased, as it is only a matter of time and labor to make plane mirrors of sufficient size. With a large plane mirror, twenty inches or upwards in size, mounted on a heliostat mounting, and so arranged as to reflect sunlight into a series of instruments rigidly supported in a horizontal position, the difficulties of eclipse observers will be very considerably lessened. The one driving clock will keep the pencil of light constantly in the same direction, and this can be used partly for photographing the corona, partly for spectroscopic work, partly for polariscopic observations, and so on for any other purposes, the whole of the instruments being fixed in the best possible positions for the observers. Practically, with a large flat mounted in the manner indicated, we can fix any portion of the sky we require to observe, and to do it we can point as many instruments as we can crowd into the pencil, each instrument being quite independent of the others. The length of focus of an objective would not introduce any difficulties on this plan, for the length of the tube is of little importance when it can be fixed in an horizontal position. The observers at the Eclipse Station only have one astronomical adjustment to make, *i.e.*, that of the position of the heliostat, and only one driving clock to regulate. This clock, since it has only to move the weight of the plane mirror and its mounting, can be more accurately made and regulated than is possible with a clock when it has to carry the weight of the tube and heavy axis of an ordinary telescope. The positions of the observers are more easy and natural during the precious seconds of totality; or, if personal superintendence is to be abolished in favor of the American pneumatic apparatus, this suggested arrangement of the instrument is better fitted for the pneumatic attachments than the old plan is. The whole of the photographic apparatus can be fixed up in a dark hut or under a dark tent with far less trouble and risk of stray light than is possible with the old manner of mounting.

The cost of a good heliostat mounting is about the same as that of a good telescope, and with one heliostat we can do the work of at least half a dozen of the usual instruments.

A. A. COMMON, F.R.S., and A. TAYLOR, A.R.S.M.

THE Supreme Court at St. Paul, Minnesota, handed down a decision on July 1st in the case of Ida Moore, of Minneapolis, against Photographer Rugg. Rugg sold a copy of Mrs. Moore's picture, which was put on exhibition in improper places, much to the discredit of the lady, and she brought suit for damages. The Supreme Court holds that it is a case in which there is ground for the recovery of damages; that the photographer has no right to dispose of pictures which are the sole property of the sitter. The decision is an important one. Similar cases have arisen once or twice previously in other parts of the country.

GENERAL NOTES.

A NOVEL CARBON PROCESS.--Mr. O. Volkner publishes the following dust carbon printing process, which appears to be easy to carry out, requires no reversed negatives, and yields permanent prints. We also think it can be used in making phototypic printing blocks. Make a solution of gelatine in water 1--60, and draw sheets of good strong paper through it, and hang it up to dry. Wet it again and squeeze it down on a piece of glass. Now brush over it a solution of ten parts gelatine, ten parts gum arabic, twenty parts white sugar, eighty parts distilled water. While still quite moist put it in a dusting box (such as used for photogravure) which contains a mixture of 100 parts of white dry sugar and five parts of French lamp black. After a lapse of eight to ten minutes, withdraw it and you will find it covered with innumerable particles of dust. Paper thus prepared will keep, and has to be sensitized in a bath of fifty parts bichromate of potassium, fifty parts bichromate of ammonia, six thousand parts water and aqua ammonia, until it assumes a light yellow color, and at last, to avoid the too quick dissolution of the gum arabic, immerse in twenty parts chromic acid in 1,500 alcohol. Print by Vogel's photometer 16° to 18°. To develop, use warm water first, and afterwards cold, leaving the print for several hours in water, to which may be added a little aqua ammonia, in case the printing was carried too far. The prints show a singular and very pleasing grain and need no transferring.--*Dr. Eder's Jahrbuch.*

EVERY photographer is, no doubt, to his own sorrow, familiar with a yellow stain in the negative, caused by taking the plate from the fixing bath before it is thoroughly fixed. Mr. Belitski, the well-known photo-chemist, made some experiments recently to remove this stain, and succeeded very well. A slight stain can often be removed by placing the negative in the

following solution: 50 parts alum, 1000 parts water, 10 parts bichromate of potassium, 20 parts muriatic acid. After several minutes the negative turns yellow all through. It is washed now very thoroughly, exposed to sunlight for several minutes, and developed or blackened with the ordinary iron developer. When the stain is very intense this remedy will not prove to be of any avail, and only by leaving it for twenty-four hours in the Lainer acid fixing bath (so often described in all journals recently) he succeeded in removing the stain and saving a valuable negative.--*Deutsche Photographen Zeitung*.

THE article on "Fixing Plates" which appeared in our June number, page 163 should read in the heading "Fixing Prints." Our readers have doubtless discovered this for themselves before now.



TRANSCRIBER'S NOTES

1. Silently corrected typographical errors and variations in spelling.
2. Retained anachronistic, non-standard, and uncertain spellings as printed.

*** END OF THE PROJECT GUTENBERG EBOOK THE AMERICAN
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